STAT443 Project Report

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1 Summary/Abstract

1.1 Project Goal

The S&P 500 is an equity market index that proxies the performance of the US market. This index plays a critical role in investment decision-making for retail and institutional investors, as well as policy makers. Forecasting the S&P 500 can help an investor determine asset allocation strategies and hedge against downside risk when major market movements are anticipated. On the other hand, economic policy makers use the S&P 500 as a leading indicator of economic expansion and contraction for interest rate cuts and hikes. Overall, we hope to find and analyze some of the most important predictors of the index.

1.2 Main Results

Through a forward selection process, we found that some of the most important predictors were silver market prices, the credit spread on Baa corporate bonds, and the CBOE Volatility Index (VIX). Combining these three variables resulted in an ARMAX model with the lowest holdout RMSE of 4.15% compared to the best baseline, exponential smoothing, and ARIMA model (**Table 1**). Other variables such as industrial production, gold prices, copper prices, and CPI were considered but rejected during forward selection.

Although the ARMAX class performed best, the IID, simple exponential smoothing (SES), and ARIMA(0,0,0) models performed relatively well. These models produced very similar holdout RMSEs of 4.33%, 4.32%, and 4.33%, respectively. We are not surprised. The S&P 500's returns are often characterized as white noise, which is what the IID/ARIMA(0,0,0) model attempt to represent. Moreover, SES's estimated smoothing parameter $\hat{\alpha}$ was very close to 0. Thus, predictions were roughly constant over time with SES.

	Persistence	**IID**	**SES**	**ARIMA(0,0,0)**	**ARMAX**
Holdout RMSE	6.49%	4.33%	4.32%	4.33%	4.15%

Table 1: Holdout RMSE

We are not surprised that ARMAX was the best. Market efficiency arguments state that no additional information about future S&P returns can be obtained from past S&P returns. External variables are integral to predicting the index, which is confirmed by our results.

Nonetheless, a RMSE of 4.15% is inadequate at face value. The IQR of S&P 500 returns is only 5.45%, meaning that the best predictions we have are still quite inaccurate. It is possible that improvements can be made to our best ARMAX model by considering other explanatory variables. For future research, some variables could include US real GDP and unemployment rate, valuation multiples (e.g., cyclically adjusted P/E ratios) to indicate under/over-valuation, and data related to investor allocations. Further gains may come from addressing outliers (e.g., the 2008 financial crisis), as such events could have distorted our model estimates and reduced predictive accuracy.

2 Variables and EDA

2.1 Forecast Variable

The variable being forecast is S&P 500 monthly returns. Let p_t be the price of the S&P 500 index at the start of month t. Then, the percentage return in month t is defined as $y_t = 100 \cdot [(p_{t+1} - p_t)/p_t]$.

We will use all returns from February 1990 to April 2015 as our training set. Observations from May 2015 to January 2026 will be in the holdout set.

To diagnose for stationarity, we created a time series plot, ACF plot, PACF plot, and 12-lag variogram of the returns data.

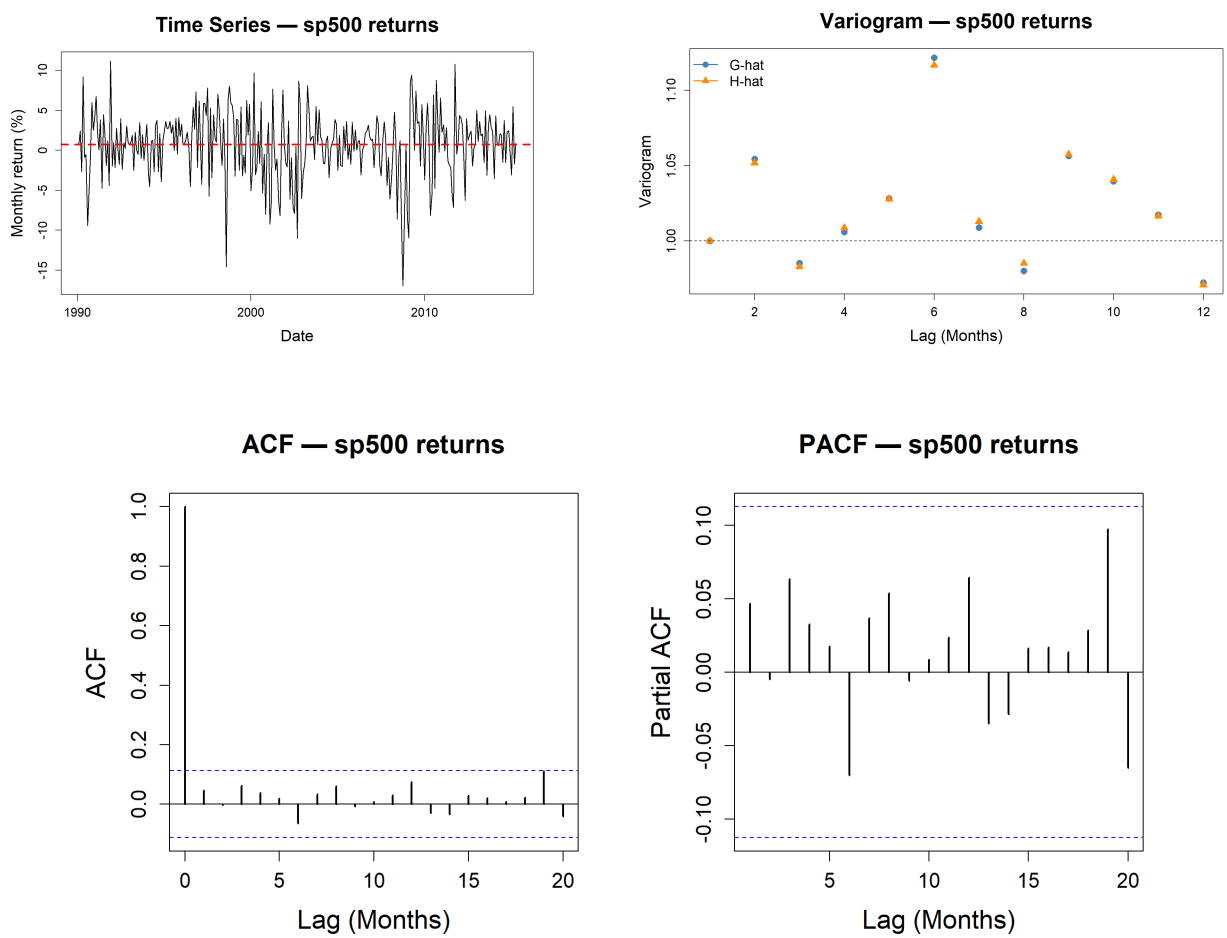


Figure 1: S&P 500 return analysis. Top: Time series and variogram of the monthly returns. Bottom: ACF and PACF plots of the monthly returns.

The time series plot in **Figure 1** demonstrates that S&P 500 returns fluctuate around a mean (red line). There is no evidence of a trend or seasonality. These findings are further supported by the variogram, ACF plot, and PACF plot in the same figure. We see that $\hat{G}_k \approx \hat{H}_k$ in the first 12 months of the variogram. Moreover, the ACF and PACF plots do not show any significant spikes. These 3 figures demonstrate that the data can be treated as a realization of a white noise process. Consequently, further predictive power should come from exogenous variables. This motivates the use of ARMAX models, which allows us to integrate other macroeconomic and financial indicators.

2.2 Base Variables

The following macroeconomic and financial variables were considered. These variables were chosen to capture different channels through which broader economic conditions and market sentiment may affect equity returns. In particular, we consider measures of real economic activity, inflation, market volatility, commodity markets, and credit conditions.

- **Industrial Production** (IP_t): A monthly index measuring the real output of the manufacturing, mining, and electric/gas utility sectors in the US during month t. It is expressed as a percentage of the production level during the base period, 2017.
- **Month over Month % Change in Consumer Price Index** (CPI_t): CPI is a monthly inflation related measure, reflecting changes in prices paid by consumers for a fixed basket of goods and services.
- **CBOE Volatility Index** (VIX_t): The market's expectation of near term S&P 500 volatility in month t, expressed in percentage terms. Higher values of VIX_t generally reflect greater uncertainty in financial markets.
- **Copper Prices** (C_t): The market price of copper in USD per ounce observed at the start of month t. Copper is often viewed as a proxy for industrial demand and broader economic activity.
- **Gold Prices** (G_t): The market price of gold in USD per ounce observed at the start of month t. Gold is commonly regarded as a safe haven asset and is often negatively correlated with equity returns.
- **Silver Prices** (AG_t): The market price of silver in USD per ounce observed at the start of month t. Like gold, silver is a precious metal, but it also has broader industrial uses that may make its relationship with equity returns different from gold.
- **Baa Credit Spread** (S_t): The difference between the yield on 20+ year Baa rated corporate bonds and the 10 year Treasury constant maturity rate in month t expressed in percentage points. This variable captures changes in perceived credit risk and financial stress.

2.3 Variable Transformations

The base variables were transformed to conduct lead-lag analysis and identify potential leading variables for an ARMAX model on S&P 500 returns. The transformed variables are:

- Difference in Industrial Production ($IP_t' = IP_t - IP_{t-1}$)
- Difference in CBOE Volatility Index ($VIX_t' = VIX_t - VIX_{t-1}$)
- Log Differenced Copper Prices ($C_t' = \log(C_t) - \log(C_{t-1})$)
- Log Differenced Silver Prices ($AG_t' = \log(AG_t) - \log(AG_{t-1})$)
- Log Differenced Gold Prices ($G_t' = \log(G_t) - \log(G_{t-1})$)
- Difference in Baa Credit Spread ($S_t' = S_t - S_{t-1}$)

Since the S&P is measured in percentage changes, we want our predictors to also be in percentage changes. Hence, we chose to difference industrial production, VIX, and credit spread, and log difference copper, silver, and gold prices.

3 Models

3.1 Baseline

The forecast equations for the baseline models are as follows.

- Persistence: $\hat{y}_{t+1|t} = y_t$
- IID: $\hat{y}_{t+1|t} = \hat{\mu}$.

For the IID model, the constant used to make predictions was $\hat{\mu} = 0.70$, which was the mean of the training set.

According to **Table 1**, the IID model has a relatively lower RMSE, which suggests that S&P 500 returns lack momentum. In other words, shocks in the data tend to disappear quickly. This result also suggests that the returns could be approximated by a white noise process, which is consistent with our analysis of the ACF and PACF plots in **Section 2.1**.

3.2 Exponential Smoothing

The forecast equation for simple exponential smoothing (SES) is

$$\hat{y}_{t+1|t} = \hat{\alpha} y_t + (1 - \hat{\alpha})\hat{\ell}_t,$$

where the smoothing parameter is estimated to be $\hat{\alpha} = 6.61 \times 10^{-5}$.

Between SES and the Holt linear model, SES had the lower holdout RMSE. This result is justified by our analysis in **Section 2.1**, as the series can be treated as a realization of a stationary process and exhibits no temporal dependence nor a seasonal pattern.

Since $\hat{\alpha}$ is very small, the model places negligible weight on the most recent observation. Hence, it follows that

$$\hat{y}_{n+h|n+h-1} \approx \hat{\ell}_{n+h-1} = \hat{\alpha} y_{n+h-1} + (1-\hat{\alpha})\hat{\ell}_{n+h-2} \approx \hat{\ell}_{n+h-2} = \hat{y}_{n+h-1|n+h-2}$$

for all $h > 0$. Hence, the 1-step forecasts are about the same throughout the holdout set.

3.3 ARIMA

The `auto.arima()` function was fitted to the training data for S&P 500 returns and the model selected was ARIMA(0,0,0), i.e. white noise

$$\hat{y}_{t+1|t} = \hat{\mu}$$

with $\hat{\mu} = 0.70$.

This result aligns with the ACF and PACF plots shown in **Section 2.1**, which show insignificant ACF and PACF for all lags. A theoretical result called weak market efficiency suggests that all past trading information, specifically historical prices and returns, cannot be used to produce superior risk-adjusted returns, which also supports a white noise model for returns.

To confirm that ARIMA(0,0,0) was the best model, p and q were perturbed and 5 alternate ARIMA models were fit with parameters $(p,d,q) \in \{(0,0,1),(0,0,2),(1,0,1),(1,0,0),(2,0,0)\}$.

Overall, the ARIMA(0,0,0) model outperformed all of the above models based on holdout RMSE and AIC. Details on all the ARIMA models can be found in **Appendix A**.

3.4 ARMAX

Before fitting the ARMAX model, CCF plots of the transformed variables with the S&P 500 were plotted to assess variables with leading effects. Moreover, we looked at variables with strong lag 0 correlation with the S&P because they could be forecast within the holdout set.

Based on the CCF values in **Table 2**, the variables used to forecast S&P 500 returns at time $t+1$ were G'_{t-9}, CPI_{t-3}, IP'_{t-1}, C'_{t-1}, AG'_t, VIX'_{t+1}, and S'_{t+1}. The CCF plots are provided in **Appendix B**.

	G'	CPI	IP'	C'	AG'	VIX'	S'
Lag	-10	-4	-2	-2	-4	0	0
CCF	0.14	0.14	0.21	0.14	0.15	0.59	0.42

Table 2: Highest absolute CCF values of each variable

We chose the first lag for AG' because the CCF values were only marginally different. Specifically, the cross-correlation was 0.137 for lag 1 and 0.148 for lag 4. Thus, we took a lag that was more recent. Since VIX' and S' at time $t+1$ are used to forecast S&P 500 returns at time $t+1$, we need to forecast those variables through the holdout set to avoid look-ahead bias.

To forecast VIX'_{t+1} and S'_{t+1}, an ARMA model was used for both variables as both series looked roughly stationary according to the time series plots below.

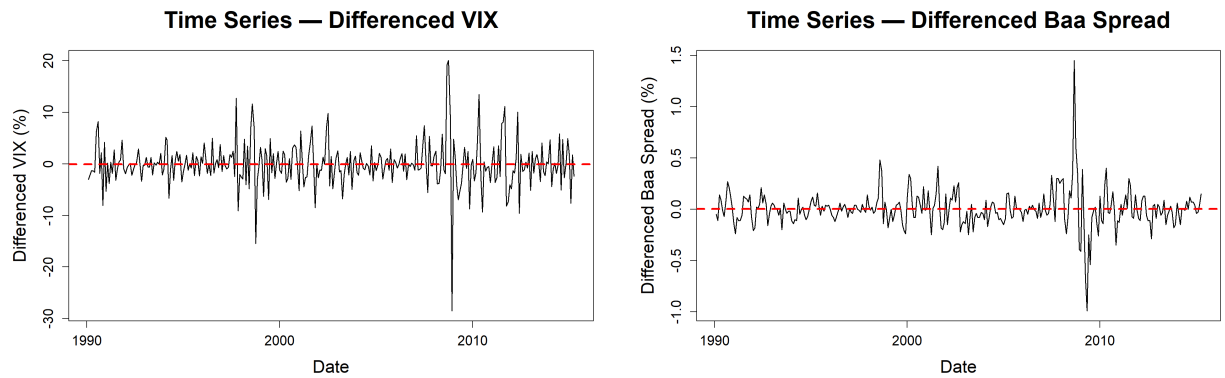


Figure 2: Time series plots of differenced Baa and differenced VIX

The model fitted by `auto.arima()` for VIX' and S' was ARIMA(0,0,3) and ARIMA(2,0,1) respectively. After forecasting these variables, forward selection was carried out to identify a good model. The selection threshold was a 1% relative reduction in holdout RMSE, i.e., a variable was included only if it reduced RMSE by at least 1% compared to the current model.

The chosen variables were arranged in descending order of their absolute correlation coefficients with the S&P 500 return series. The first variable in the list (i.e., VIX') was added to the model and the holdout RMSE was computed for this after fitting the ARMAX model.

Finally, we iterate through the remaining variables in the order specified earlier using the following algorithm:

1. Add the selected variable to the model and refit it over training data.
2. Compute the holdout RMSE of the newly fitted model.
3. If the holdout RMSE improved by at least 1% relative to the existing model, the variable is kept in the model permanently. If not, the variable is discarded.

Steps 1-3 are repeated until all variables are used up. The forecast equation of the best model based on this process was

$$\hat{y}_{t+1|t} = \hat{\beta}_0 + \hat{\beta}_1 \widehat{VIX'}_{t+1|t} + \hat{\beta}_2 \widehat{S'}_{t+1|t} + \hat{\beta}_3 AG'_t.$$

Notice that the residual process in the model follows white noise. Furthermore, the coefficient estimates are $\hat{\beta}_0 = 0.70,\ \hat{\beta}_1 = -0.47,\ \hat{\beta}_2 = -4.22$, and $\hat{\beta}_3 = 2.42$.

The coefficient values obtained align with macroeconomic intuition. Higher values of VIX generally signal market stress, panic selling and investor fear, which is associated with lower S&P 500 returns. Higher values of the Baa spread also indicate rising investor anxiety and worsening economic conditions, which is linked to lower S&P 500 returns. Silver has significant industrial applications, which is why positive silver returns are associated with economic expansions, and higher S&P 500 returns.

4 Contributions

4.1 How the Team Formed

Samuel knew most group members from a variety of past courses he has taken. Tai was from MATH 320, Sarah was from STAT 305, and Chakradhar was from MATH 316. Miguel joined our group later in the term based on Harry's recommendation.

4.2 Ordering of Names

The authors are listed alphabetical by last name

4.3 Major Contributions

- Tai: Exploratory data analysis and visualization
- Sarah: Report writing, project infrastructure/reproducibility
- Samuel: Team leader, organized meetings, report writing, proofreading
- Miguel: Report writing
- Chakradhar: Data preprocessing, model building

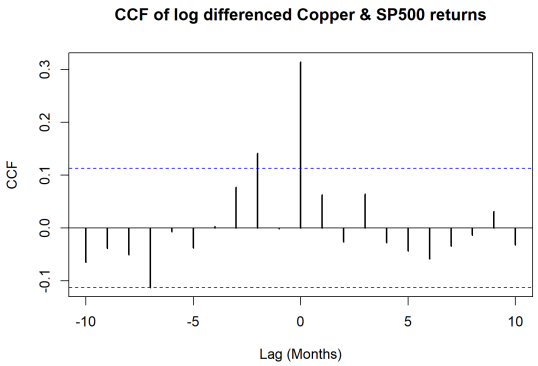
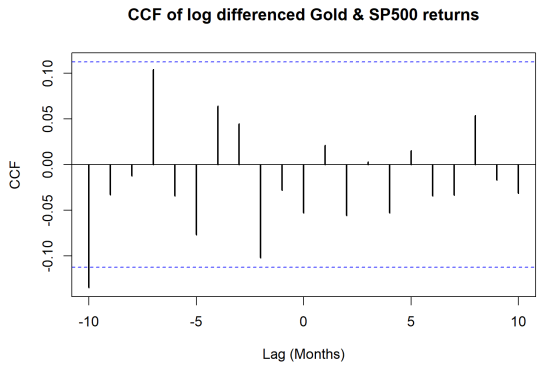
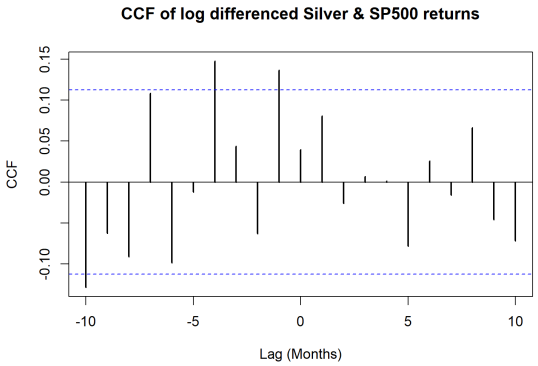
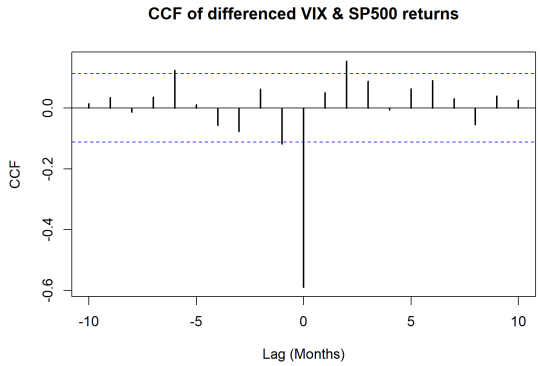
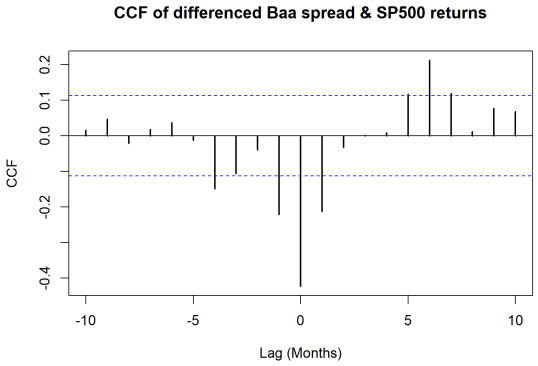
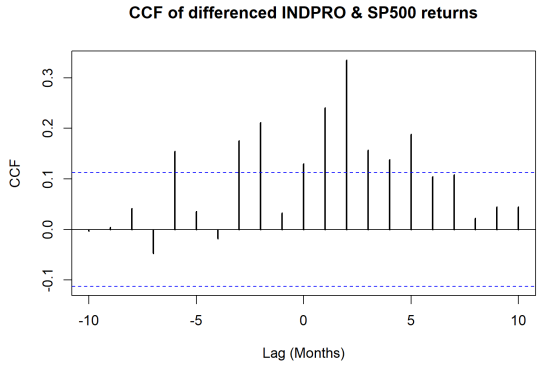
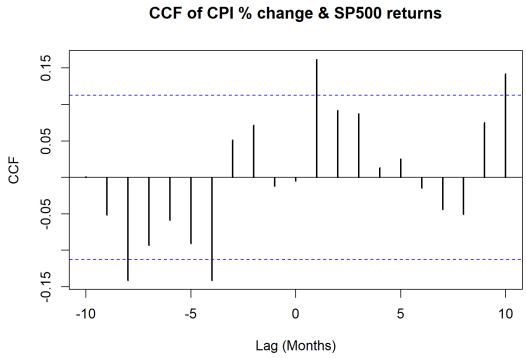
5 Appendix

A Model AIC & Holdout RMSE

	(0,0,0)	**(0,0,1)**	**(0,0,2)**	**(1,0,0)**	**(1,0,1)**	**(2,0,0)**
AIC	1731.38	1732.71	1734.69	1732.72	1734.24	1734.72
Holdout RMSE	4.327%	4.356%	4.352%	4.356%	4.331%	4.354%

Table 1: Statistics of ARIMA models

	RMSE
$\widehat{VIX'}_{t+1\|t}$	4.31%
$\widehat{VIX'}_{t+1\|t}+\widehat{S'}_{t+1\|t}$	4.20%
$\widehat{VIX'}_{t+1\|t}+\widehat{S'}_{t+1\|t}+IP'_{t-1}$	4.57%
$\widehat{VIX'}_{t+1\|t}+\widehat{S'}_{t+1\|t}+CPI_{t-3}$	4.20%
$\widehat{VIX'}_{t+1\|t}+\widehat{S'}_{t+1\|t}+C'_{t-1}$	4.20%
$\widehat{VIX'}_{t+1\|t}+\widehat{S'}_{t+1\|t}+AG'_t$	4.15%
$\widehat{VIX'}_{t+1\|t}+\widehat{S'}_{t+1\|t}+AG'_t+G'_{t-9}$	4.13%

Table 2: Holdout RMSE of ARMAX models

B CCF Plots of Predictors